

RQ Answers

omitted

2024-04-05

Data overview

glimpse(all_data)				
## Rows: 812				
## Columns: 11				
## \$ repo	<chr> "cayenne", "cayenne", "cayenne", "cayenne", "cayenne..."			
## \$ pr_number	<int> 511, 524, 524, 524, 524, 524, 524, 529, 529, 529, ...			
## \$ rule	<chr> "java:S1192", "java:S1192", "java:S1192", "java:S1192", "java:S1192", ...			
## \$ severity	<chr> "CRITICAL", "CRITICAL", "CRITICAL", "CRITICAL", "CRITICAL", ...			
## \$ file	<chr> "cayenne-server/src/test/java/org/apache/cayenne/c..."			
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", ...			
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", ...			
## \$ debt	<int> 8, 10, 12, 10, 10, 5, 8, 14, 18, 14, 2, 14, 2, 14, 2, 2, ...			
## \$ ncloc_affected_file	<int> 58, 329, 329, 329, 329, 329, 329, 329, 243, 243, 243, 2, ...			
## \$ complexity	<int> 3, 26, 26, 26, 26, 26, 26, 26, 18, 18, 18, 18, 18, 18, ...			
## \$ origin	<chr> "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", ...			

RQ1

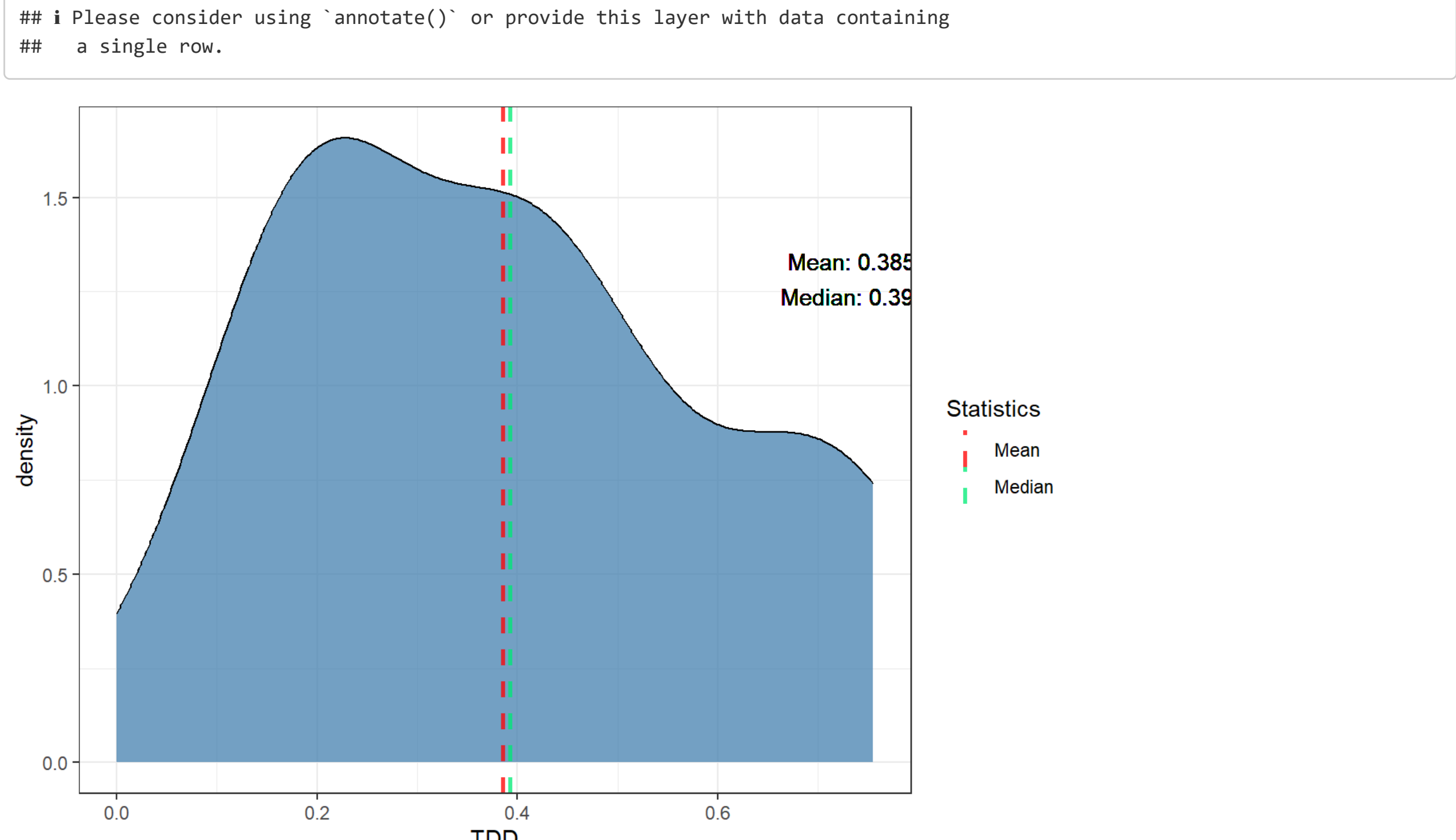
TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd)), : All aesthetics have length 1, but the data has 27 rows.
```

```
## # Please consider using `annotate()` or provide this layer with data containing a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd)), : All aesthetics have length 1, but the data has 27 rows.
```

```
## # Please consider using `annotate()` or provide this layer with data containing a single row.
```



TDV (Technical Debt Variation)

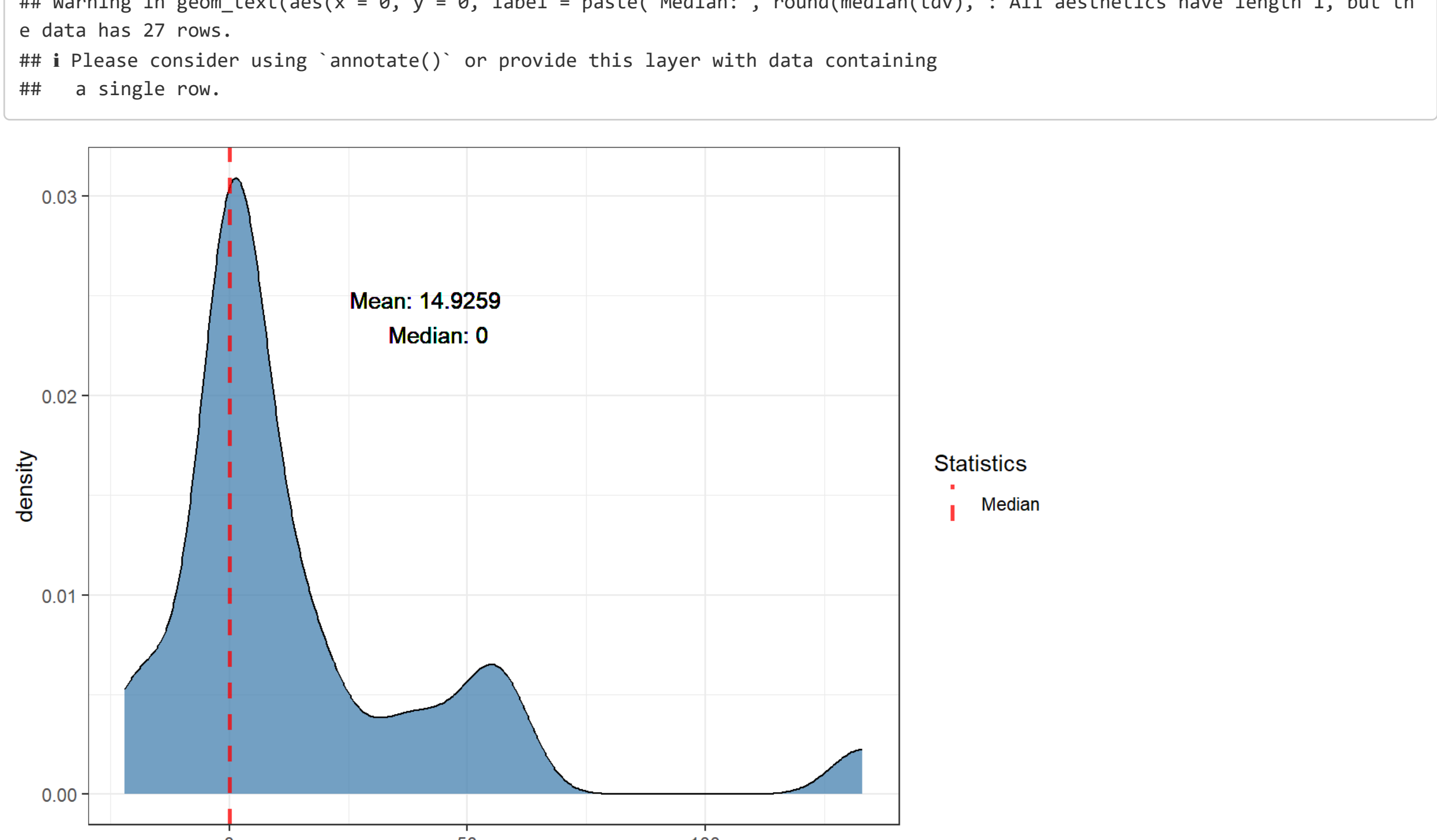
TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv)), : All aesthetics have length 1, but the data has 27 rows.
```

```
## # Please consider using `annotate()` or provide this layer with data containing a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv)), : All aesthetics have length 1, but the data has 27 rows.
```

```
## # Please consider using `annotate()` or provide this layer with data containing a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 cayenne	11.1	40.7	48.2	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	<dbl>
## 1	11.1	40.7	48.2

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>
## 1 cayenne	68	32	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	68	32

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 cayenne	93.3	6.67	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	93.3	6.67

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 9 × 2		
## Groups: rank [9]		
## rank cayenne		
## <int> <chr>		
## 1	1 java:S1192	- 7
## 2	2 java:S1874	- 6
## 3	3 java:S3740	- 2
## 4	4 java:S1301	- 2
## 5	5 java:S108	- 1
## 6	6 java:S100	- 1
## 7	7 java:S2293	- 1
## 8	8 java:S2142	- 1
## 9	9 java:S1124	- 1

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 9 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1192	122	
## 2 java:S1874	123	
## 3 java:S1301	124	
## 4 java:S3740	125	
## 5 java:S100	126	
## 6 java:S108	127	
## 7 java:S1124	128	
## 8 java:S2142	129	
## 9 java:S2293	130	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## Groups: rank [10]		
## rank cayenne		
## <int> <chr>		
## 1	1 java:S1192	- 368
## 2	2 java:S1845	- 59
## 3	3 java:S101	- 54
## 4	4 java:S100	- 47
## 5	5 java:S2692	- 32
## 6	6 java:S1301	- 28
## 7	7 java:S1659	- 26
## 8	8 java:S117	- 24
## 9	9 java:S112	- 16
## 10	10 java:S2111	- 12

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1192	122	
## 2 java:S1845	123	
## 3 java:S101	124	
## 4 java:S100	125	
## 5 java:S2692	126	
## 6 java:S1301	127	
## 7 java:S1659	128	
## 8 java:S117	129	
## 9 java:S112	130	
## 10 java:S2111	131	

Outliers

Higher TDD

## # A tibble: 27 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 cayenne	557	450	596	0.755
## 2 cayenne	541	236	313	0.754
## 3 cayenne	565	539	729	0.739
## 4 cayenne	563	238	348	0.684
## 5 cayenne	576	962	1456	0.661
## 6 cayenne	604	531	843	0.630
## 7 cayenne	556	1285	2479	0.518
## 8 cayenne	544	560	1157	0.484
## 9 cayenne	547	577	1243	0.464
## 10 cayenne	537	32	75	0.427
## # i 17 more rows				

Higher TDV

## # A tibble: 27 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	<dbl>
## 1	563 cayenne	133	
## 2	567 cayenne	58	
## 3	544 cayenne	57	
## 4	556 cayenne	52	
## 5	558 cayenne	42	
## 6	604 cayenne	34	
## 7	565 cayenne	22	
## 8	588 cayenne	16	
## 9	596 cayenne	16	
## 10	535 cayenne	10	
## # i 17 more rows			

Lower TDV

## # A tibble: 27 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	<dbl>
## 1	541 cayenne	-22	
## 2	569 cayenne	-18	
## 3	529 cayenne	-14	
## 4	524 cayenne	0	
## 5	537 cayenne	0	
## 6	538 cayenne	0	
## 7	539 cayenne	0	
## 8	542 cayenne	0	
## 9	547 cayenne	0	
## 10	557 cayenne	0	
## # i 17 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 25	

Number of java:S1192 issues that affect test code

## n	
## 1 375	

Number of java:S1192 issues that affect not test code

## n	
## 1 0	

Summary NCLOC

## ncloc	
## Min.:	58.0
## 1st Qu.:	172.0
## Median:	414.0
## Mean:	694.6
## 3rd Qu.:	1189.5
## Max.:	2330.0